



THE POLYTECHNIC
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF COMPUTING AND INFORMATION
TECHNOLOGY
Bachelor of Mathematics and Statistics
[MTS]
Year 2

End of Year Examinations
Computer Graphics and Applications
(MTS-CGA-221)

22nd September, 2021	Room: CAFE	Examiner: R. Banda	Time Allowed: 3 Hrs
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Instructions

- *There are **FOUR** Questions and each question carries 25 marks.*
- *Answer **ALL** Questions.*
- *Begin each question on a fresh page.*
- *Show all your work clearly.*

[DO NOT TURNOVER UNTIL INSTRUCTED TO DO SO]

Question one**25 Marks**

- a. Define the following terms: **5 marks**
- Computer graphics
 - Shadow mask
 - Noninterlaced system
 - Rasterization
 - Framebuffer depth
- b. The four major areas of computer graphics are: display of information, user interfaces, design and simulation. With an example, describe how each area is applied **4 marks**
- c. How do you understand computer graphics? **2 marks**
- d. With the help of examples, describe the difference between painting program and drawing program. **2 marks**
- e. Describe two parts which distinguish computer graphics system from ordinary computer system **2 marks**
- f. Explain the importance of double buffering in animation **2 marks**
- g. Explain two properties of a raster images **2 marks**
- h. Describe anti-aliasing **2 marks**
- i. With the help of examples, describe the difference between lossless data compression and lossy data compression. **2 marks**
- j. Explain the differences between GKS and PHIGS standards **2 marks**

Question two**25 Marks**

- a. Describe the color model used in computer graphics **2 marks**
- b. Explain alpha blending in computer graphics **2 marks**
- c. With examples, explain the three basic kinds of geometric transform **3 marks**
- d. Distinguish stroke and fill ways of drawing a shape **2 marks**

- e. Describe the role homogeneous coordinates in computer graphics **2 marks**
- f. With the aid of examples, explain the three OpenGL graphics libraries **3 marks**
- g. Differentiate between object coordinates and eye coordinates **2 marks**
- h. Explain what would happen to the program if you omit `#include <GL/gl.h>` and `#include <GL/glut.h>` **2 marks**
- i. State what happens at each of the following statements: **5 marks**
 - i. `glFlush();`
 - ii. `glutInit(&argc, argv);`
 - iii. `glutInitDisplayMode(GLUT_RGB);`
 - iv. `glutDisplayFunc(display);`
 - v. `glutMainLoop ();`
- j. Why is OpenGL called a state machine **2 marks**

Question three **25 Marks**

- a. Define a computer graphics rendering pipeline **1 mark**
- b. Outline the main stages in 3D graphics pipeline **2 marks**
- c. Describe clipping in the context of OpenGL, and where does it fit in the steps of the OpenGL pipeline architecture? **2 marks**
- d. Describe OpenGL depth test **2 marks**
- e. Differentiate material and texture **2 marks**
- f. Use diagrams only, illustrate the following:
 - i. Perspective projection and Orthographic projection **2 marks**
 - ii. Window and Viewport **2 marks**

- g. Differentiate between illumination and shading **2 marks**
- h. Explain the difference between a directional light and a point source light in the context of OpenGL **2 marks**
- i. In our illumination model, there were three terms: an ambient term, a diffuse term, and a specular term.
 - i. What effect does the ambient term model? **2 marks**
 - ii. What assumption(s) do we make for the diffuse term? **2 marks**
 - iii. In the specular term, there is a factor of $(r \cdot v)^{sp}$. What effect does varying the specular power (sp) have? **2 marks**
- j. State two general standards developed for graphics software **2 marks**

Question four **25 Marks**

- a. Define a digital image **1 mark**
- b. State the first two fundamental steps in image processing **2 marks**
- c. Describe three application areas of image processing with appropriate examples. **3 marks**
- d. With the help of a diagram, explain the digital image acquisition process **4 marks**
- e. Explain any two simplest image enhancement techniques **2 marks**
- f. Explain three specialized image processing hardware **3 marks**
- g. With the aid of a diagram, illustrate how a model of the image degradation / restoration process occurs **4 Marks**
- h. Differentiate the illumination and reflectance of a digital image formation **2 marks**
- i. Distinguish the following terms

- i. Sampling and quantization **2 marks**
- ii. spatial domain and frequency domain methods. **2 marks**

The End



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of BUSINESS AND APPLIED SCIENCES

SCHOOL OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS
Bachelor of Mathematics and Statistics in Education
[MTS]
Year 2

End of Semester Examinations
Computer Graphics and Applications
(MTS-CGA-221)

25 October, 2023

Examiner: R. Banda

Time Allowed: 3 Hours

Instructions

- There are **FOUR** Questions and each question carries 25 marks.
- Answer **ALL** Questions.
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Question one **25 Marks**

- a. Write brief notes of the following:
 - i. Modern computer graphics challenges **3 marks**
 - ii. Computer resolution **3 marks**
 - iii. View transformation **3 marks**
- b. The four major areas of computer graphics are: display of information, user interfaces, design and simulation. With an example, describe how each area is used **4 marks**
- c. Describe two parts which distinguish computer graphics system from ordinary computer system **4 marks**
- d. Explain two properties of a raster images **2 marks**
- e. Differentiate between parallel projection and perspective projection. **2 marks**
- f. With the help of examples, explain the difference between lossless data compression and lossy data compression. **4 marks**

Question two **25 Marks**

- a. Describe the color model used in OpenGL computer graphics **2 marks**
- b. Explain alpha blending in computer graphics **2 marks**
- c. With the aid of examples, explain the three OpenGL graphics libraries **3 marks**
- d. Describe any two characteristics of frame buffer that affect graphics performance **4 marks**
- e. Outline the main stages in graphics pipeline **4 marks**
- f. Describe OpenGL state machine **2 marks**
- g. Differentiate between illumination and shading **2 marks**
- h. In our illumination model, there were three terms: an ambient term, a diffuse term, and a specular term.
 - i. What effect does the ambient term model? **2 marks**
 - ii. What assumption(s) do we make for the diffuse term? **2 marks**

20/02/2020

- iii. In the specular term, there is a factor of $(r \cdot v)^{sp}$. What effect does varying the specular power (sp) have? **2 marks**

Question three **25 Marks**

- a. Explain the purpose of transformations in computer graphics **2 marks**
- b. State any role of matrices in computer graphics **1 mark**
- c. Write Matrix of 2D Rotation **2 marks**
- d. With examples on each, explain the three basic classes of 3D transformations **3 marks**
- e. Describe the transform needed to transform the triangle from A to B in Figure 1 assuming that the origin is placed in the lower left corner of triangle A. **4 marks**

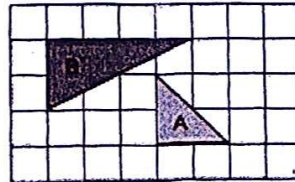


Figure 1: Transform from A to B

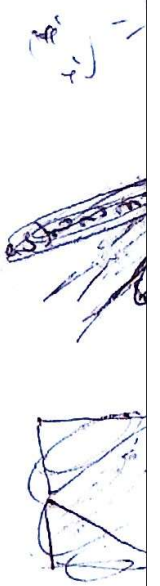
- f. Using diagram only, illustrate Perspective Projection and Orthographic Projection **4 marks**
- g. Define the following terms:
- i. Interpolation **1 mark**
 - ii. View frustum culling **1 mark**
 - iii. Viewport **1 mark**
- h. How do you understand Pattern recognition? **2 marks**
- i. Explain any two machine vision applications **4 marks**

Question four

25 Marks

- a. Describe two application areas of image processing with appropriate examples.
4 marks
- b. With an illustration, explain the Image acquisition process **4 marks**
- c. With the aid of a diagram, illustrate how a model of the image degradation / restoration process occurs
4 Marks
- d. Differentiate the illumination and reflectance of a digital image formation
2 marks
- e. Distinguish the following terms
- i. Sampling and quantization **2 marks**
 - ii. spatial domain and frequency domain methods. **2 marks**
- f. Discuss sampling artifacts and their effect on image quality on a raster display.
5 marks
- g. What can be done to reduce or eliminate them?
2 marks

The End



Bachelor of Mathematics and Statistics
[MTS]
Year 2
MID SEMESTER TEST

Computer Graphics and Application COM 208

INSTRUCTION(S): Answer ALL questions

1. Define the following: **(1 mark each)**
 - a. Computer graphics
 - b. refresh rate
 - c. rasterization
 - d. Pixel
 - e. frame-buffer
2. With examples distinguish the following **(2 marks each)**
 - a. window port & view port
 - b. Raster and vector images
 - c. RGBA and HSLA
 - d. Local and infinite light sources
 - e. ambient reflection and specular reflection
3. Discuss the three shading models used in computer graphics **(9 marks)**
4. With the aid of examples, explain the three OpenGL graphics libraries **(6 marks)**
5. Explain three OpenGL basic transformations **(6 marks)** *translation, rotation, scaling*
6. Use Figure 1 attached to answer the following questions:
 - i. Why is it important to include `#include<windows.h>` and `#include<GL/glut.h>` libraries? **(2 marks)**
 - ii. Explain each statement in the main function **(1 mark each)**
 - iii. Describe the role of each of these gl functions in the display function:
`glClearColor ( ), glClear ( ), glBegin ( ), glColor3f ( ), glVertex2f ( ), glEnd ( ), glFlush ( )`

```

1  #include<windows.h>
2  #include<GL/glut.h>
3  void display() {
4      glClearColor(0.0f, 0.0f, 0.0f, 1.0f);
5      glClear(GL_COLOR_BUFFER_BIT);
6
7      glBegin(GL_TRIANGLE_STRIP);
8          glColor3f(1.0f, 0.0f, 0.0f);
9          glVertex2f(-0.5f, -0.5f);
10         glVertex2f(0.5f, -0.5f);
11         glVertex2f(0.5f, 0.5f);
12         glVertex2f(-0.5f, 0.5f);
13     glEnd();
14     glFlush();
15 }
16 int main(int argc, char** argv) {
17     glutInit(&argc, argv);
18     glutCreateWindow("OpenGL Setup Test");
19     glutInitWindowSize(320, 320);
20     glutInitWindowPosition(50, 50);
21     glutDisplayFunc(display);
22     glutMainLoop();
23     return 0;
24 }

```

Figure 1: OpenGL program

[MSE]

Year 2

COM-208 COMPUTER GRAPHICS AND APPLICATIONS
TEST III

Time: 1hr : 30 mins

INSTRUCTIONS:

- i. Answer ALL Questions
- ii. Write neatly and legibly

COMPUTER GRAPHICS AND APPLICATIONS 50 MARKS

Question 1 : Illumination and shading

- i. Describe the primary goal of illumination and shading in computer graphics? **2 Marks**
- ii. Explain the difference between global and local illumination. **4 Marks**
- iii. Explain Lambertian reflection, and how does it affect shading? **4 Marks**
- iv. Describe ambient lighting and its role in shading. **4 Marks**
- v. Explain Phong shading and its advantages over Gouraud shading. **4 Marks**
- vi. Briefly describe the three components of Phong reflection model? **6 Marks**

Question 2: Pattern Recognition

- i. Describe pattern recognition and explain its significance in computer science and engineering. **4 Marks**
- ii. Differentiate between classification and clustering in pattern recognition. **4 Marks**
- iii. Explain the concept of support vector machines (SVM) in pattern recognition and their role in binary classification. **4 Marks**

Question 3: Digital Image processing

- i. Define digital image processing and explain its significance in various fields. **4 Marks**
- ii. What is the fundamental difference between analog and digital images? **2 Marks**
- iii. Describe the process of image acquisition and its role in digital image processing. **4 Marks**
- iv. Explain the concept of pixel and its significance in digital images. **4 Marks**



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- c. Describe two parts which distinguish computer graphics system from ordinary computer system **4 marks**
- d. Explain two properties of a raster images **4 marks**
- e. With the help of examples, explain the difference between lossless data compression and lossy data compression. **4 marks**

Question two **25 Marks**

- a. Describe the color model used in OpenGL computer graphics **2 marks**
- b. Explain alpha blending in computer graphics **2 marks**
- c. With examples, explain the three basic kinds of geometric transform used in OpenGL **3 marks**
- d. With the aid of examples, explain the three OpenGL graphics libraries **3 marks**
- e. Describe any two characteristics of frame buffer that affect graphics performance **4 marks**
- f. Define a computer graphics rendering pipeline **1 mark**
- g. Outline the main stages in 3D graphics pipeline **4 marks**
- h. Describe OpenGL depth test **2 marks**
- i. Differentiate between illumination and shading **2 marks**
- j. State two general standards developed for graphics software **2 marks**

Question three**25 Marks**

- a) How do you understand Pattern recognition? **2 marks**
- b) Describe the difference between Statistical pattern recognition and Structural pattern recognition **4 marks** *analisis*
- c) Describe two key ideas on which Support Vector Machines learning algorithms are based **4 marks**
- d) Differentiate between Convolutional neural networks (CNNs) and Recurrent neural networks (RNNs) **4 marks** *over*
- e) State any three classifiers used in computer vision **3 marks** *memor*
- f) Describe two strengths of neural networks learning **4 marks** *Artificial*
- g) Explain any two machine vision applications **4 marks** *Robot*
drum.

Question four**25 Marks**

- a. Define a digital image **1 mark**
- b. State the first two fundamental steps in image processing **2 marks**
- c. Describe three application areas of image processing with appropriate examples. **6 marks**
- d. Explain the Image acquisition process **6 marks**
- e. With the aid of a diagram, illustrate how a model of the image degradation / restoration process occurs **4 Marks**
- f. Differentiate the illumination and reflectance of a digital image formation **2 marks**
- g. Distinguish the following terms
 - i. Sampling and quantization **2 marks**
 - ii. spatial domain and frequency domain methods. **2 marks**

The End